

SPRINT 1 — DATA FOUNDATION

Nifty 100 Financial Intelligence Platform

Technical Documentation

1. Executive Summary

- The Data Foundation sprint ingested 12 distinct Excel files into a 1.49 MB SQLite database (`nifty100.db`).
- Achieved a 100% test pass rate (43/43 tests) and enforced 0 foreign key violations across 92 validated companies.
- The final validation run produced zero CRITICAL findings.

2. Sprint Objective

Consolidate financial datasets, normalize identifiers, enforce referential integrity, apply DQ-01 to DQ-16 validation rules, and establish a robust baseline for analytics.

3. Scope & Deliverables

Verified presence of 9/9 required artifacts:

#	Artifact
1	nifty100.db
2	load_audit.csv
3	validation_failures.csv
4	loader.py
5	normaliser.py
6	validator.py
7	schema.sql
8	tests/etl/
9	exploratory_queries.sql

Submission folders: Source Code/, Datasets/, Documentation/, PPT Slides/, Demo Video/.

4. Source Data

Ingested 12 raw Excel files mapping to business entities:

analysis.xlsx	market_cap.xlsx
balancesheet.xlsx	peer_groups.xlsx
cashflow.xlsx	profitandloss.xlsx
companies.xlsx	prosandcons.xlsx
documents.xlsx	sectors.xlsx
financial_ratios.xlsx	stock_prices.xlsx

5. ETL Architecture

12 Raw Excel Sources
 -> Data Normalisation
 -> ETL Loader
 -> Deduplication + FK Filtering
 -> SQLite Database
 -> Data Quality Validation
 -> Audit + Validation Reports

Core scripts executed: normaliser.py, loader.py, validator.py, schema.sql.

- loader.py: dropped duplicate primary-key intersections and pre-filtered rows containing unlisted company tickers to protect database integrity.
- validator.py: executed post-load; evaluated accounting rules (e.g., DQ-04 Assets vs Liabilities) and generated the final validation_failures.csv.

6. Normalisation

normaliser.py applied programmatic manipulations to clean Ticker IDs and cast disparate Fiscal Year strings into a unified YYYY format.

7. Database Schema

14 physical tables are currently present in SQLite, including downstream Sprint 2/Sprint 3 tables and the SQLite internal sequence table. The original Sprint 1 schema contains the required base tables.

Group	Tables
Master	companies, sectors
Financial	profitandloss, balancesheet, cashflow
Market / Supporting	stock_prices, market_cap, analysis, documents, prosandcons, peer_groups
Downstream / Internal	financial_ratios, peer_percentiles, sqlite_sequence

8. Data Quality Framework

- Validation rules DQ-01 to DQ-16 are applied post-load by `validator.py`.
- Findings are graded CRITICAL, WARNING or INFO and written to the audit outputs rather than silently discarded.
- The ETL layer rejected fatal data prior to insertion, so no CRITICAL findings reached the database.

9. Validation Results

Severity	Count	Notes
CRITICAL	0	ETL layer rejected fatal data prior to insertion
WARNING	319	Logical edge cases, e.g. DQ-16 missing 5-year coverage
INFO	1,175	Positive alignments
Total	1,494	Across 92 distinct companies

Note: 92 companies with findings means every company generated at least one warning/info log, not that they failed.

10. Automated Testing

43 out of 43 Pytest ETL tests passed (100% success rate).

Collected	Passed	Failed	Skipped	Pass rate
43	43	0	0	100%

11. Database Statistics

Table	Rows	Note
companies	92	
profitandloss	1,070	
balancesheet	1,140	
cashflow	1,056	
analysis	4	
documents	1,456	
prosandcons	14	
sectors	92	
stock_prices	5,520	
market_cap	552	
peer_groups	56	
financial_ratios	1,070	Downstream / Sprint 2 protected data
peer_percentiles	560	Downstream / Sprint 3 generated
sqlite_sequence	2	SQLite system internal

12. Implementation Decisions & Deviations

- **Ratio preservation:** rather than overwriting the database with the raw `financial_ratios.xlsx`, the ETL protected 1,070 rows of pre-computed Sprint 2 logic to prevent regressions in downstream workflows.
- **Table variances:** the database holds 14 tables rather than the original 12. This accommodates SQLite's internal sequence tracker and future-proofs Sprint 3 analytics (`peer_percentiles`).

13. Known Data Limitations

Source raw files contained a small number of orphaned tickers (companies listed in financial files but absent from the master sheet). These were rejected by the pipeline to ensure 0 foreign key violations.

14. Final Verification

Check	Result
PRAGMA foreign_key_check	0 violations
PRAGMA integrity_check	ok
Required artifacts	9/9
ETL tests	43/43 passed
CRITICAL DQ findings	0

15. Sprint Closure

Sprint 1 is officially categorized as **Completed / Ready to Close**.